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Seven million years of wind and precipitation variability on the Chinese Loess Plateau

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ARTICLE INFO

Article history:

Received 25 December 2009

Received in revised form 23 June 2010

Accepted 4 July 2010

Available online 24 July 2010

Editor: P. DeMenocal

Keywords:

Chinese Loess Plateau

wind

precipitation

East Asian monsoon

ABSTRACT

Recent studies of the eolian deposits in northern China have extended the East Asian monsoon history back to the early Miocene. However, the relative intensities of monsoonal winds and precipitation and the extent of their coupling prior to the Pleistocene epoch remain poorly constrained, mainly due to uncertainties in the interpretation of proxy indices generated from the Mio–Pliocene Red Clay sequences. Here we reconstruct East Asian monsoon oscillations over the past 7 Ma using magnetic susceptibility and carbonate content as summer monsoon (precipitation) proxies, and quartz grain size as a winter monsoon (wind intensity) index. Our results suggest that precipitation and wind intensity exhibited significant orbital-scale variations prior to 4.2 Ma, followed by slightly damped variability between 4.2 and 2.75 Ma. Subsequently, East Asian monsoon circulation experienced two large shifts at about 2.75 and 1.25 Ma, characterized by stepwise strengthening of glacial wind and interglacial precipitation. A remarkable change in East Asian monsoon seasonality occurred around 3.15–2.75 Ma. Prior to 3.15 Ma, strong winds were positively correlated with high effective precipitation, whereas after 2.75 Ma strong winds were negatively correlated with heavy precipitation. This shift was probably induced by a change from an insolation-forced system to one strongly influenced by the combined effects of the phased uplift of the Himalaya–Tibetan Plateau and the simultaneous development of the Northern Hemisphere ice sheets.

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1. Introduction

Eolian deposits in northern China consist of the Quaternary loess–paleosol sequences underlain by the late Neogene Red Clay formation (e.g., Liu, 1985; Ding et al., 1998; Sun et al., 1998; An, 2000; Guo et al., 2002). Over the past two decades, much work has focused on this unique continental archive with the goal of understanding the history and variability of East Asian monsoon circulation (e.g., An et al., 1990; Liu and Ding, 1998; An, 2000; Guo et al., 2002). Both geological evidence and climate-model simulations suggest that large shifts of the East Asian monsoon since the Miocene were probably associated with the phased uplift of the Himalaya–Tibetan Plateau (e.g., An et al., 1990; Xiao and An, 1999; An et al., 2001), whilst monsoon variability at orbital-to-millennial time scales is dynamically linked to changes in external solar insolation and internal boundary conditions such as ice volume and ocean–atmosphere energy exchange (e.g., Prell and Kutzbach, 1987; An et al., 1991a; Liu and Ding, 1993; Ding et al., 1995; An, 2000; Vandenberghe et al., 2006; Clemens et al., 2008).

East Asian monsoon circulation is characterized by significant seasonal changes in wind vigor and precipitation, and both produce

distinctive signals in Chinese loess (An et al., 1990; Liu and Ding, 1998; An, 2000). For example, strengthened summer precipitation results in enhanced magnetic mineral content during pedogenesis (e.g., Zhou et al., 1990; Heller and Evans, 1995; Maher and Thompson, 1995). In contrast, intensified winter or spring winds transport large amounts of eolian dust which is deposited mostly within the Chinese Loess Plateau (An et al., 1991b; Xiao et al., 1995; Sun et al., 2001; Nugteren and Vandenberghe, 2004; Prins et al., 2007). Loess-based proxies such as magnetic susceptibility and particle size have been widely used to evaluate monsoon variability on tectonic to millennial timescales (Kukla, 1987; Kukla and An, 1989; An et al., 1990, 1991a,b; Bloemendal et al., 1995; Porter and An, 1995; Liu et al., 1999).

The alternating loess–paleosol sequences, characterized by large-amplitude fluctuations of both magnetic susceptibility and grain size, clearly document an enhanced summer monsoon during Pleistocene interglacials and a strengthened winter monsoon during Pleistocene glacials (e.g., Kukla and An, 1989; An et al., 1990; Heller and Evans, 1995; Maher and Thompson, 1995; Liu and Ding, 1998; Xiao and An, 1999; An, 2000). The underlying Pliocene Red Clay formation, however, exhibits relatively low amplitude orbital-scale oscillations of both magnetic susceptibility ($\sim 78 \times 10^{-8} \text{ m}^3 \text{ kg}^{-1}$) and bulk grain size ($< 3 \mu\text{m}$ for mean grain size) (An et al., 1999; Ding et al., 1999; An, 2000; Ding et al., 2000). Previous studies have tried to decipher the palaeoclimatic system during the period of the Red

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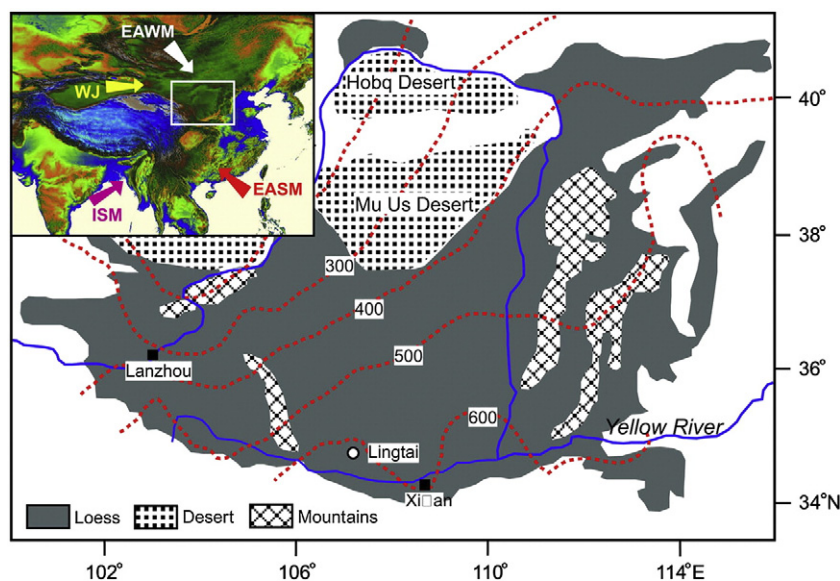


Fig. 1. Map showing major atmospheric circulation regimes in East Asia (imbedded) and the Chinese Loess Plateau with the location of the Lingtai section. Dashed lines denote contours of mean annual precipitation (mm). ISM—Indian summer monsoon, EASM—East Asian summer monsoon, EAWM—East Asian winter monsoon, WJ—Westerly jet.

Clay formation (An et al., 1999; Ding et al., 1999; Guo et al., 2002; Vandenberghe et al., 2004); however the trend and variability of East Asian monsoon circulation during the Mio–Pliocene are still not well understood, mainly due to inconsistent interpretations of these proxy indicators. For example, bulk grain size of the Red Clay has been interpreted as an indicator of westerly circulation (Ding et al., 1998), winter monsoon intensity (An, 2000; An et al., 2001) or combinations thereof (Vandenberghe et al., 2004; Sun et al., 2008). A schematic map of these wind systems is shown in Fig. 1. Similarly, interpretation of Red Clay magnetic susceptibility as an indicator of summer monsoon strength remains controversial (Sun et al., 1998; An et al., 2001; Ding et al., 2001a; Liu et al., 2003).

To achieve a more thorough understanding of East Asian monsoon circulation since the late Miocene, records of quartz grain size, magnetic susceptibility and carbonate content were generated from a continuous loess–paleosol and Red Clay sequence in the Chinese Loess Plateau. Inter-proxy comparisons indicate that carbonate content and magnetic susceptibility are reliable proxies for effective precipitation, although orbital-scale magnetic susceptibility variations are damped in the Red Clay formation. Unlike previous suggestions of less variable monsoon fluctuations during the late Miocene to Pliocene, our results reveal that orbital-scale oscillations of both wind and precipitation were persistently significant over the last 7 million years. Moreover, the coupling mode of the precipitation and wind intensity differed before and after 2.75 Ma, implying a significant role of changing lower boundary conditions (e.g., regional tectonic events and onset of the Northern Hemisphere Glaciation) in modulating the long-term change of East Asian monsoon circulation.

2. Materials and methods

The loess–paleosol and Red Clay sequence investigated in this study is located at Lingtai (35°04'N, 107°39'E, 1350 m above sea level) on the southern Loess Plateau (Fig. 1). At present, the mean annual temperature and precipitation at Lingtai are about 8.8 °C and 650 mm. The field outcrop consists of the Quaternary loess–paleosol sequence (168 m) underlain by the 120-m-thick late Neogene Red Clay formation (Fig. 2). Descriptions of the pedomorphology and magnetostratigraphy of the Lingtai section were detailed in previous studies (Sun et al., 1998; Ding et al., 1999). In the field, samples were taken at 8 cm intervals for the loess–paleosol sequences and 4 cm intervals

for the Red Clay formation. A total of 5100 samples were collected for proxy analyses.

Magnetic susceptibility was measured using a Bartington MS2 meter at the Institute of Earth Environment, Chinese Academy of Sciences. Carbonate content was determined by a volumetric method, with an error of less than 3%. For grain size analysis, all the samples were pretreated by adding 30% hydrogen peroxide (H_2O_2) and 6 N hydrochloric acid (HCl) to remove organic matter, carbonate, and iron oxides. Monomineralic quartz particles were then isolated using potassium pyrosulfate ($K_2S_2O_7$) fusion and hydrofluorosilicic acid (H_2SiF_6) (Xiao et al., 1995; Sun et al., 2000). Meanwhile, 720 samples were pretreated by only removing organic matter, carbonate and iron oxides (hereafter referred to as bulk samples). Particle-size distributions of bulk and quartz samples were determined using a Malvern Mastersize S laser-diffraction analyzer at the Institute of Earth Environment, Chinese Academy of Sciences.

The chronological framework of the loess–paleosol and Red Clay sequence was generated by palaeomagnetic studies (Sun et al., 1998; Ding et al., 1999), and subsequently refined for the last 3.6 Ma by using an orbital tuning approach (for a detailed description, see Sun et al., 2006a). Prior to 3.6 Ma, the age of each sample from the Lingtai section was calculated by linear interpolation between the age points derived from the geomagnetic reversals (Sun et al., 1998). The chronology of the Lingtai section presented in this study is composed of two parts: an orbitally tuned age model for the last 3.6 Ma and a linearly interpolated time scale for the interval 7–3.6 Ma. Estimated from this chronology, the sedimentation rate (SR) varies from 1.3 to 7 cm/ka for the Late Neogene Red Clay sequence and from 1.5 to 37.3 cm/ka for the Quaternary loess–paleosol sequence (Fig. 2). Our sampling strategy yields a resolution of ≤ 3 kyr for the entire eolian sequence. Significant SR changes, as indicated by the averaged SR and glacial–interglacial SR variability, imply that remarkable shifts of both Asian interior drying and dust transport dynamics may have occurred around 2.7 Ma (Ding et al., 2000; Sun and An, 2005).

The carbonate content of the Lingtai section is reported for the first time in this paper. The results of quartz grain size and magnetic susceptibility measurements are extended to the late Miocene with a much higher resolution than used in previous work (Sun et al., 2006a,b). Since the three proxies were generated from the same depth, the phase and coupling relationships among these proxies discussed in the following sections are intrinsic and

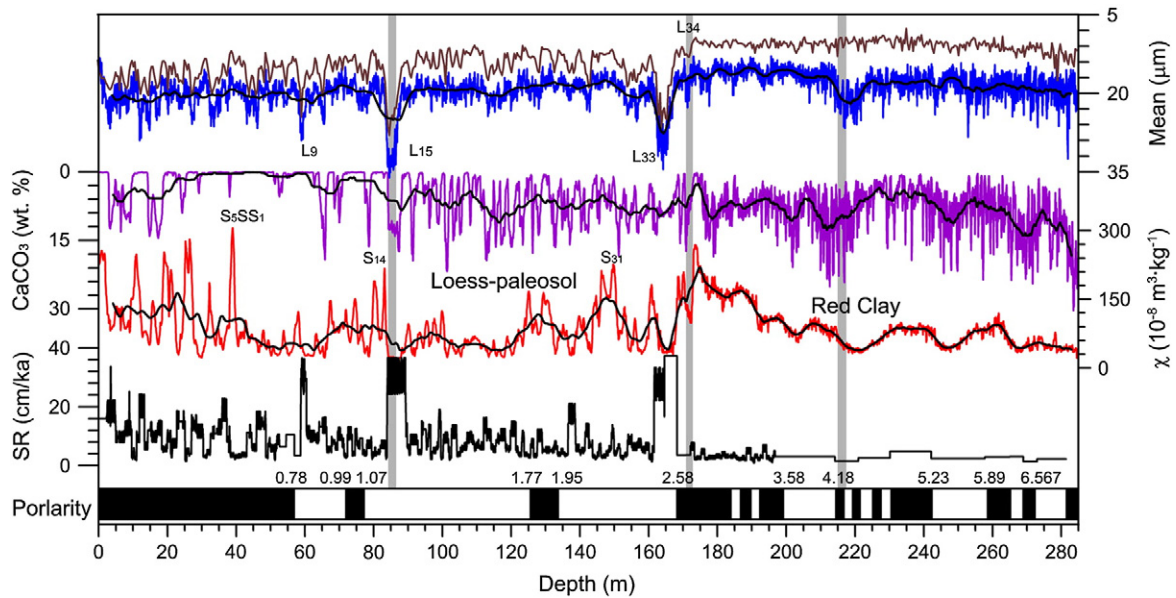


Fig. 2. Magnetostratigraphy and changes in sedimentation rate (SR), magnetic susceptibility (χ), carbonate content, and mean grain size of bulk (black) and quartz (blue) samples from the Lingtai section. Magnetostratigraphy results are derived from previous studies (Sun et al., 1998). Heavy lines denote the long-term trends of the magnetic susceptibility, carbonate content and quartz grain size. Grey bars denote apparent shifts in the amplitude of the three proxies.

independent of the chronology. Therefore, our results permit a robust evaluation of orbital-scale variations of the wind vigor and precipitation since the late Miocene.

3. Paleoclimatic implications of quartz grain size, carbonate content and magnetic susceptibility

To ascertain the advantage of quartz grain size as a reliable indicator of the wind strength (Xiao et al., 1995), we compare mean grain size variations of the bulk and quartz samples through time (Fig. 2). In the loess–paleosol sequences, the mean grain size of both bulk and quartz samples exhibits similar orbital-scale variability, although the mean grain size of quartz particles is relatively coarse compared to the bulk samples. For the Red Clay formation, the mean grain size of bulk samples shows little variability, whereas the mean grain size of quartz particles exhibits oscillations very similar to those of the overlying loess–paleosol sequences. Moreover, the mean grain size difference between the bulk and quartz samples is larger in the Red Clay formation than in the overlying loess–paleosol sequences. These differences suggest that pedogenic modification of the non-quartz components is stronger in the Red Clay sequence than in the overlying loess–paleosol deposits.

While the particle size of eolian deposits might be influenced by factors such as the source area extent and/or aridity, source-sink distance and paleowind intensity (An et al., 1991b; Xiao et al., 1992; Rea, 1994; Ding et al., 2001b), comparison of the quartz mean grain size (hereafter referred to as QGS) with dust flux suggests that the aridity in inland Asia and paleowind intensity were weakly related prior to 2.7 Ma, but became gradually coupled over the last 2.7 Ma (Sun and An, 2005). There has been no significant change in the provenance of the Red Clay deposit (Sun, 2005; Wang et al., 2007; Sun and Zhu, 2010), and therefore the Pleistocene interpretation of the QGS as an indicator of the wind strength (Xiao et al., 1995) can be extended into the Pliocene and Miocene (Sun et al., 2006a,b). Fluctuations of the QGS can be divided into four stages, as indicated by three remarkable coarsening events around 220 m, 165 m (L_{33}), and 85 m (L_{15}) (Fig. 2). The first stage (220–285 m) is characterized by relatively strong wind strength and significant variability, whereas in the second interval both the averaged wind intensity and orbital-

scale variability are greatly damped. Subsequently, the averaged intensity and glacial–interglacial variability of the winter monsoon demonstrate a stepwise increase around 165 and 85 m.

Carbonate content is a function of the soil forming environment, such as the balance between precipitation-induced leaching intensity and temperature-related evapotranspiration effectiveness (Reheis, 1987). A seasonally wet/dry climate rather than the amount of rainfall, with a critical balance between rainfall and evaporation, is considered to be a key factor in driving carbonate dissolution and reprecipitation (Rossinsky and Swart, 1993). In the monsoon-affected area (e.g., the Chinese Loess Plateau), variation in carbonate content is controlled mainly by the effective precipitation (i.e., the balance between precipitation-induced leaching and temperature-induced evaporation) (Lu, 1981; Liu, 1985). The carbonate content in Chinese loess–paleosol sequences was interpreted as a proxy indicator of the effective precipitation that is associated with the summer monsoon intensity, with relatively low values in the paleosol layers due to strong precipitation-induced leaching (An et al., 1991b). Carbonate content has been employed as a precipitation proxy to reconstruct rapid summer monsoon oscillations as well as the long-term summer monsoon evolution (Chen et al., 1997, 1999a,b; Fang et al., 1999; Chen et al., 2007).

At the Lingtai section, the carbonate content exhibits large amplitude variability (0.7–25%) in the Red Clay formation, slightly lower (0.2–21.8%) variability from L_{33} to L_{15} , and damped variability (0–19.2%) above L_{15} (Fig. 2). The primary difference between the three stages is the gradual decreases in the averaged carbonate content and the glacial–interglacial variability, suggesting stepwise strengthening of the effective precipitation particularly during the warm periods. The decreased glacial–interglacial variability of carbonate content in the upper loess–paleosol sequence (S_0 – S_{14}) was probably caused by the relatively heavy precipitation during interglacial intervals, which results in strong dissolution of carbonate not only in the paleosol layers but also in the uppermost part of the underlying loess layers. In addition, U-series dating and stable isotope analysis of carbonate nodules in Chinese loess also suggest that eluviation and reprecipitation of soil carbonates also occurred during glacial stages (Rowe and Maher, 2000).

Magnetic susceptibility is a widely employed indicator of summer monsoon intensity (An et al., 1991a; Heller and Evans, 1995; Maher and Thompson, 1995), although the relationship between magnetic susceptibility and pedogenic intensity remains controversial for the Red Clay formation (Ding et al., 1998; Sun et al., 1998; An et al., 1999; Ding et al., 2001a; Vandenberghe et al., 2004). Similar magnetic properties and a close relationship between the magnetic susceptibility and Rb/Sr ratio (an independent weathering index, Chen et al., 1999a,b) indicate that magnetic susceptibility of the Red Clay formation can provide a general understanding of long-term summer monsoon variation (An et al., 2001; Liu et al., 2003).

At the Lingtai section, the variability of magnetic susceptibility in the Red Clay differs from that of the loess–paleosol sequences in terms of both amplitude and frequency (Fig. 2). In the Red Clay formation, it displays relatively low-frequency and small-amplitude variability. Long-term variability of the magnetic susceptibility is rather uniform in the lower portion (215–285 m) but increases gradually in the upper part of the Red Clay formation (170–215 m). Magnetic susceptibility in the loess–paleosol sequences, however, exhibits large amplitude glacial–interglacial fluctuations. Moreover, the long-term trend is relatively stable during glacials but shows a sawtooth pattern during interglacials (e.g., three remarkable magnetic susceptibility strengthening around S_{31} , S_{14} and S_5SS_1 followed by gradual decreasing trends) (Fig. 2).

Visual inspection suggests that variations in carbonate content and magnetic susceptibility are different in three aspects. First, magnetic susceptibility of the Red Clay sequence can only document the low-frequency aspects of summer monsoon evolution, but is less sensitive to orbital-scale monsoonal variation. In contrast, the carbonate content exhibits distinct orbital-scale variability with a long-term trend similar to that of magnetic susceptibility. Second, in the lower loess–paleosol sequence (L_{34} – L_{15}), magnetic susceptibility captures the changing amplitude of strong summer monsoon events during the Pleistocene interglacials, whereas it is relatively uniform (20 – $45 \times 10^{-8} \text{ m}^3 \text{ kg}^{-1}$) in the weakly weathered loess layers and thus may not reflect the real intensity of weak summer monsoons. Carbonate content, however, does capture the amplitude of both strong and weak monsoon events within this depth interval (85–170 m). Third, for the upper loess–paleosol sequence (S_{14} – S_0), carbonate content is almost reduced to zero in all of the paleosol layers and the uppermost portion of most of the loess layers, probably because of leaching due to strong precipitation during interglacial times. In contrast, magnetic susceptibility exhibits distinct glacial–interglacial variability with two remarkable increases around S_{14} and S_5SS_1 superimposed.

These discrepancies are mainly due to different post-depositional processes associated with the two proxies, e.g., the magnetic susceptibility of a layer is a property of the same layer, while the carbonate is dissolved from one layer but reprecipitated in a lower

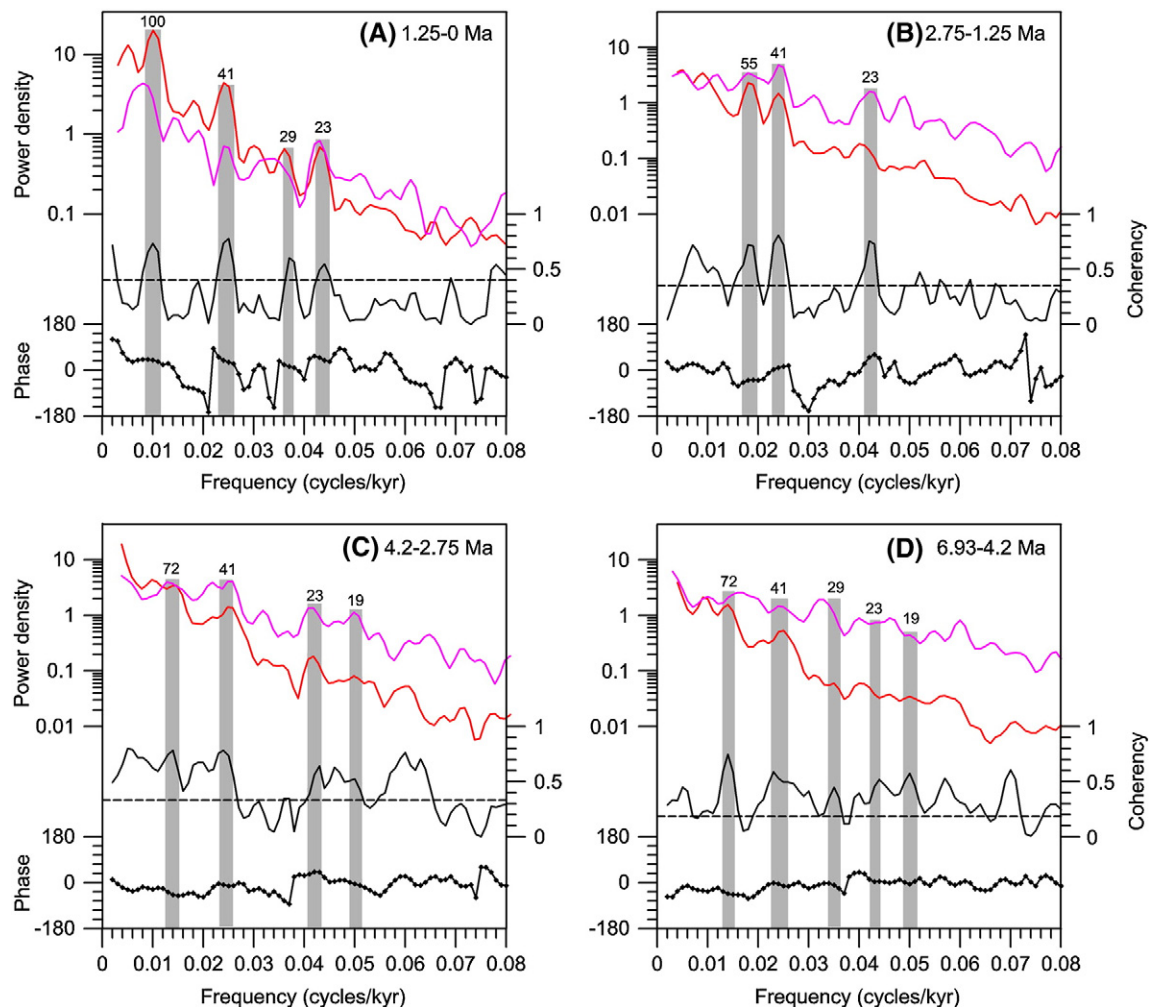


Fig. 3. Cross-spectral results of magnetic susceptibility (red) and carbonate content (pink) for four intervals: 6.93–4.2 Ma, 4.2–2.75 Ma, 2.75–1.25 Ma, and 1.25–0 Ma. Periods of the main spectral peaks above 90% confidence level are labeled. Spectral densities are plotted on log scales. The horizontal dashed lines indicate confidence at the 80% level in the coherence spectra. Grey bars denote strong coherence and near-zero phase relationships between magnetic susceptibility and carbonate content at different primary and non-primary orbital bands. 0° phase indicates that high magnetic susceptibility and low carbonate content are in-phase.

layer. Magnetic susceptibility (a physical proxy) enhancement in Chinese loess is probably associated with the formation of ultrafine ferromagnetic minerals (e.g., magnetite and maghemite) during pedogenesis under relatively warm and humid conditions (Zhou et al., 1990; Maher and Thompson, 1991; Heller et al., 1993; Evans and Heller, 1994; Heller and Evans, 1995; Maher, 1998). Previous investigation of the relationship between magnetic susceptibility and precipitation suggests that increased mean annual precipitation (up to 1200 mm) results in increased magnetic susceptibility, while the link between precipitation and magnetic susceptibility would be significantly weakened when the precipitation is less than 200 mm (Maher and Thompson, 1995; Han et al., 1996a,b). During glacial times, the magnetic properties of the weakly weathered loess units tend to be dominated by coarse-grained lithogenic magnetite, rather than pedogenic magnetite/maghemite (Deng et al., 2005). Thus, magnetic susceptibility of glacial loess layers is less sensitive to precipitation changes compared to that of interglacial paleosol units.

Carbonate content (a chemical proxy) in Chinese loess is controlled mainly by effective precipitation, with relatively low values in the paleosol layers due to strong precipitation-induced leaching (Lu, 1981; Liu, 1985; An et al., 1991b). When the precipitation ranges between 200 and 500 mm, carbonate content has a negative relationship with changing precipitation (Huang et al., 2008). However carbonate is almost dissolved when the precipitation

exceeds 600 m (e.g., Bloemendal and Liu, 2005; Chen et al., 2007) and thus carbonate content cannot reflect the true amplitude of strong summer monsoon events during peak interglacials. Visual differences between magnetic susceptibility and carbonate content suggest that in the Red Clay sequence carbonate content is a better proxy for the effective precipitation variability than magnetic susceptibility, whereas in the upper loess–paleosol sequence (S_{14} – S_0) magnetic susceptibility is a more reliable indicator of the summer monsoon strength compared to carbonate content.

While the sensitivity of magnetic susceptibility and carbonate content to precipitation variability is not similar, cyclic oscillations are readily matched between magnetic susceptibility and carbonate content. For example, in the loess–paleosol sequences, the low carbonate content corresponds well with high magnetic susceptibility during interglacials, and vice versa. Similar correlations are also identified in the Red Clay formation, although magnetic susceptibility variation is subtle and less distinct than the carbonate content. Cross-spectral results provide further evidence of the close relationship between the magnetic susceptibility and carbonate content records (Fig. 3). Over the past 1.25 Ma, both records exhibit dominant peaks around 100-, 41- and 23-kyr, and these two records are strongly coherent and almost in-phase at these primary orbital periods. During the interval 2.75–1.25 Ma, cross-spectral results of these two records show distinct peaks, high coherence, and a near-zero phase

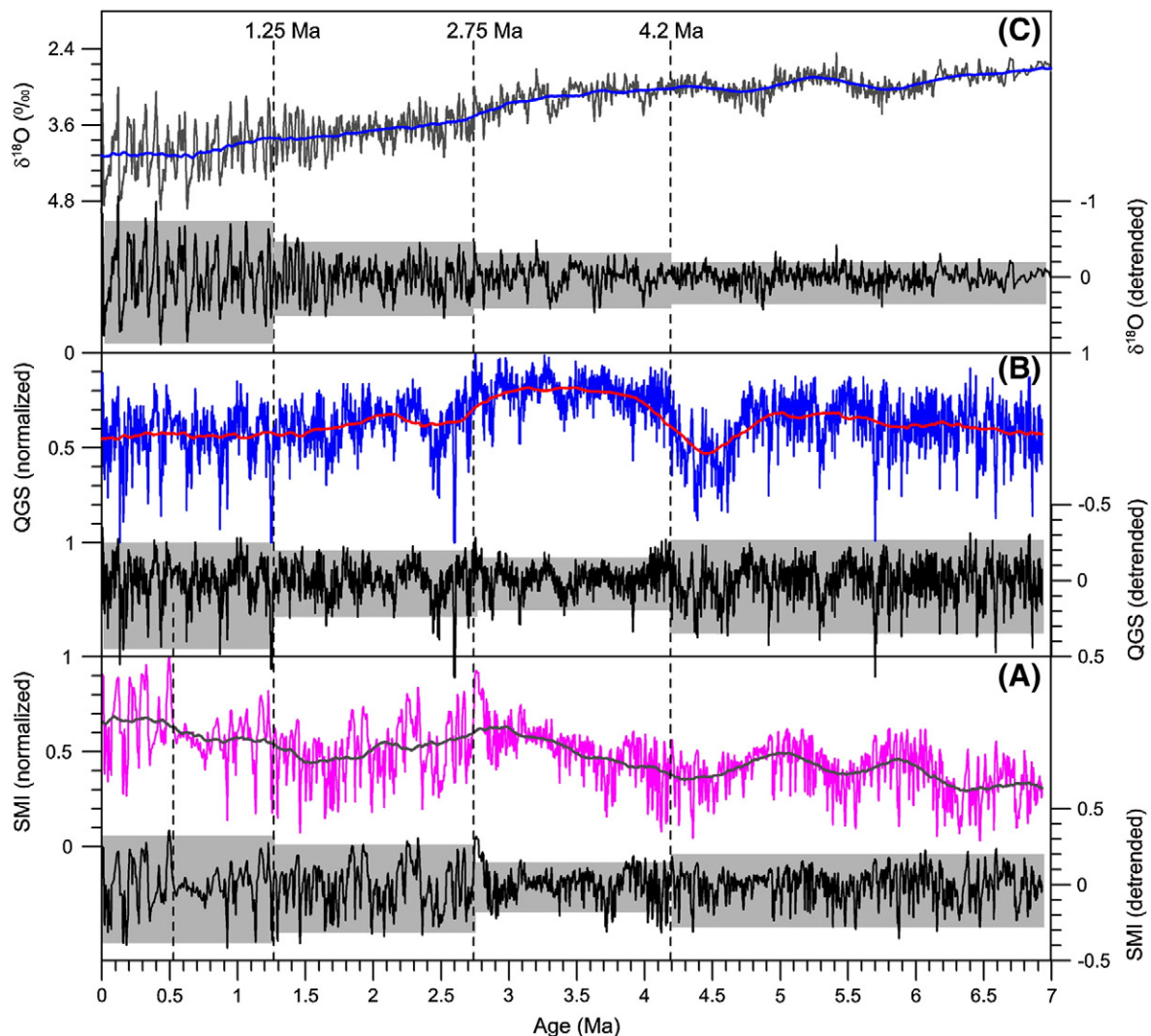


Fig. 4. Comparison of (A) SMI and (B) QGS with (C) composite $\delta^{18}\text{O}$ record (Zachos et al., 2001). Heavy lines denote the long-term mean trends of the three proxies. Also shown are the detrended results of the three proxies. Dashed lines indicate three large shifts in the amplitude of the SMI and QGS around 4.2 Ma, 2.75 Ma and 1.25 Ma.

relationship at the 41- and 23-kyr periods. Prior to 2.75 Ma, both magnetic susceptibility and carbonate content spectra exhibit peaks around 41-, 23-, and 19-kyr and high coherency with near-zero phase relationship at the obliquity and precession bands. In addition to the primary orbital peaks, non-primary peaks (i.e., 72-, 55-, and 29-kyr) are also evident in the spectrum, probably resulting from heterodyne interactions between mechanisms operating at the three orbital bands.

Given the high coherence and in-phase relationship between magnetic susceptibility and carbonate content, we combine these two records by normalizing each proxy and averaging them to produce a stacked summer monsoon index (SMI). Note that the SMI contains the distinctive variability of both proxies. For example, prior to the Pleistocene, SMI variability mainly reflects the orbital-scale carbonate variations. However, the SMI variability since 1.25 Ma is mainly attributable to glacial–interglacial magnetic susceptibility variations. Since precipitation is a major control on magnetic enhancement (Liu et al., 1995; Maher and Thompson, 1995; Han et al., 1996a,b) and carbonate leaching in Chinese loess (Liu, 1985; An et al., 1991b), we interpret the SMI as mainly indicative of precipitation variability, with

higher values indicating more effective precipitation. For convenience of comparison, the QGS is also normalized, with high values reflecting strong winter monsoon intensity (Xiao et al., 1995) and possibly also the expansion of the source area (Ding et al., 2005). We next address the trends and variability of East Asian monsoon circulation over the past 7 Ma using time series of these two proxies illustrated in Fig. 4. By applying a high-pass filter (<500 kyr), the long-term trends can be extracted from the original data sets, and the detrended record regarded as a good expression of the glacial–interglacial variability.

4. Trend and variability of East Asian monsoon circulation

The long-term mean of the SMI was relatively stable before 4.2 Ma and increased gradually during 4.2–2.75 Ma. Subsequently, the SMI mean decreased until 1.5 Ma and increased again over the last 1.5 Ma (Fig. 4A). Unlike the SMI long-term mean, the QGS mean trend suggests that the wind intensity was relatively strong before 4.2 Ma and subsequently weakened until 2.75 Ma. After 2.75 Ma, the long-term mean of the wind intensity increased gradually and became stable over the last 1.25 Ma (Fig. 4B). Based upon differences in the

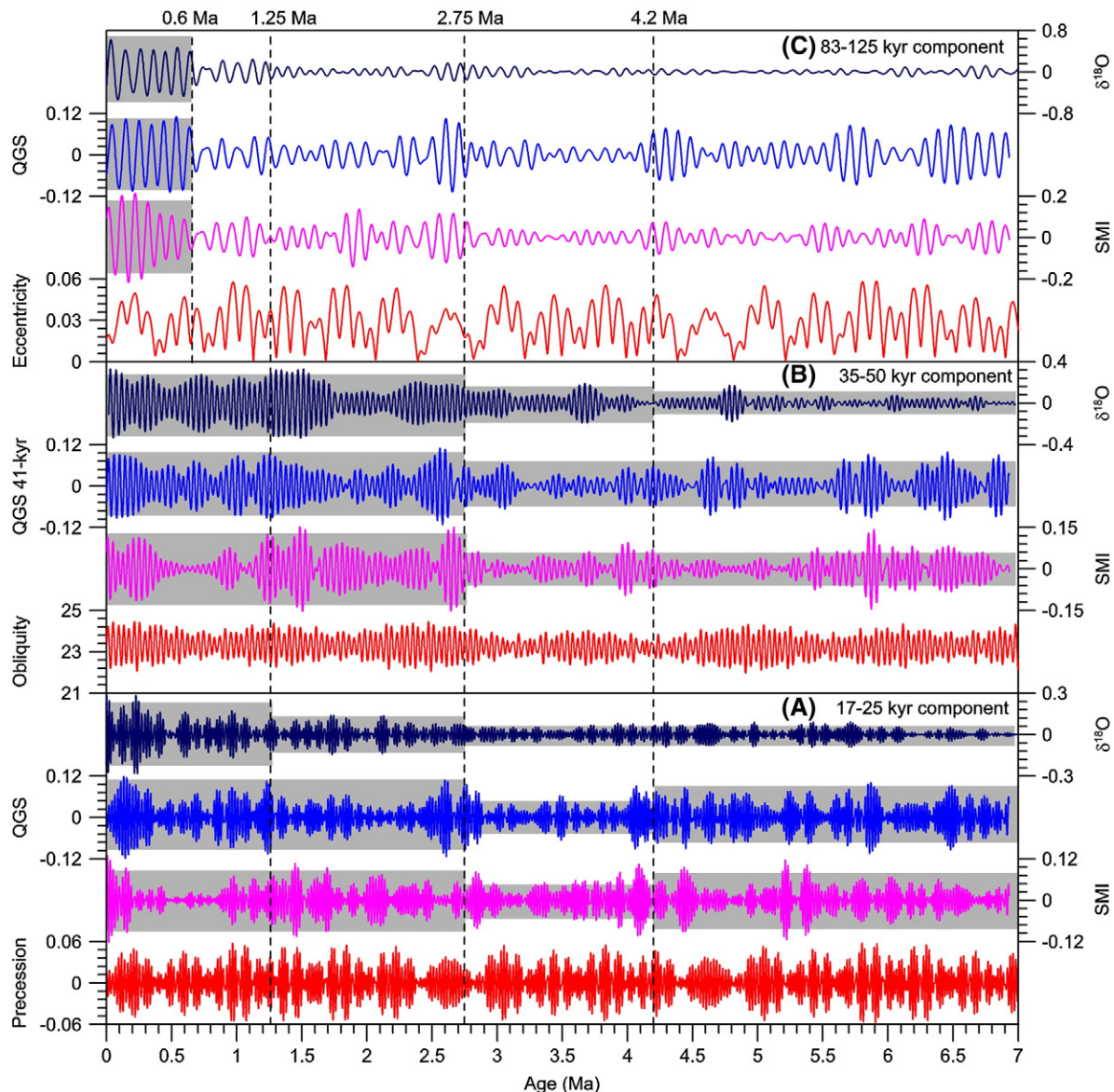


Fig. 5. Comparison of orbital parameters (i.e., eccentricity, obliquity and precession, Laskar et al., 2004) with filtered components of the SMI, QGS and $\delta^{18}\text{O}$ records (Zachos et al., 2001) at the 23-, 41-kyr, and 100-kyr bands. Grey bars denote the amplitude of filtered components of difference proxies at the three orbital bands. Dashed lines indicate three large shifts in the East Asian monsoon circulation occurred around 4.2 Ma, 2.75 Ma, and 1.25 Ma.

long-term means and glacial–interglacial amplitude of the SMI and QGS records, evolution of the wind and precipitation regimes can be divided into four intervals: (1) 7–4.2 Ma; (2) 4.2–2.75 Ma; (3) 2.75–1.25 Ma; and (4) 1.25–0 Ma. The 4.2-Ma shift is followed by gradually increasing of the precipitation (SMI), relative weakening of the wind intensity (QGS), and the simultaneously damped variability of both precipitation and wind strength. The 2.75-Ma shift corresponds with the boundary between the late Neogene Red Clay and Quaternary loess–paleosol sequence (Sun et al., 2006a; Yang and Ding, 2010). The 1.25-Ma shift is characterized by an abrupt increase in the magnetic susceptibility of paleosol S₁₄ and a gradual coarsening of the QGS of the glacial loess layers; this shift has been reported in numerous loess publications (e.g., An et al., 1990; Xiao and An, 1999; Sun and Liu, 2000; Sun et al., 2006a).

During the first interval (7–4.2 Ma), SMI variability is significant and comparable to that of the third interval (2.75–1.25 Ma), whereas the SMI amplitude was slightly damped within the second interval (4.2–2.75 Ma). Afterwards, the amplitude of the SMI variation exhibits two step-like amplifications around 2.75 Ma and 1.25 Ma (Fig. 4A). The QGS variation reveals that the wind intensity varied significantly in terms of both long-term trend and orbital-scale variability (Fig. 4B). Detrended QGS variations are also characterized by four discrete intervals with regard to amplitude: (1) moderate amplitude prior to 4.2 Ma; (2) small-amplitude between 4.2 and 2.75 Ma; (3) moderate amplitude during 2.75–1.25 Ma; and (4) large amplitude over the last 1.25 Ma.

Clearly, both the long-term mean trends and glacial–interglacial amplitude of the wind and precipitation oscillations over the past 7 Ma are different. Since external insolation does not show any amplitude shifts or long-term means over the past 7 Ma (Berger and Loutre, 1991), different trends and glacial–interglacial variability of the winter and summer monsoons are probably attributable to different responses of these two monsoon subsystems to changing lower boundary conditions, in particular global ice volume and regional tectonic events. Climate-model simulations suggest that stronger glacial conditions result in a strengthening of the winter monsoon and a weakening of the summer monsoon (Prell and Kutzbach, 1992), whereas tectonic uplift can lead to the simultaneous strengthening of the winter and summer monsoon (An et al., 2001). Detailed comparisons of paleoclimate records from land and ocean suggest that the 4.2-Ma shift might be linked to phased uplift of the Himalayan–Tibetan Plateau, since the gradual strengthening of the summer monsoon around 4.2–2.75 Ma cannot be attributed to increased ice volume (An et al., 2001). The 2.75- and 1.25-Ma shifts, however, are probably linked to the onset and full development of the Northern Hemisphere glaciation respectively (Liu and Ding, 1993). Different trends and variability of the SMI and QGS over the past 7 Ma indicate that the responses of the wind and precipitation in East Asia to tectonic uplift and ice volume change are rather complicated at tectonic to orbital time scales.

Considering the amplitude shifts of orbital-scale variability and the long-term trend difference of these two proxies, it is suggested that three significant changes in East Asian monsoon circulation took place around 4.2 Ma, 2.75 Ma and 1.25 Ma (Fig. 4). Detrended QGS and SMI results suggest that during the first interval (7–4.2 Ma) orbital-scale variability of the wind and precipitation intensity is comparable to that of the third interval (2.75–1.25 Ma), implying that significant monsoon oscillations were persistent at least since the late Miocene. This inference differs from previous suggestions that this interval can be considered as an initial stage of the East Asian paleomonsoon circulation, as revealed by insignificant variations of both bulk grain size and magnetic susceptibility (An et al., 1999; An, 2000; Ding et al., 2000). More recently, however, numerous proxies derived from the Red Clay formation, such as the free iron to total iron ratio (Ding et al., 2001a), chemical weathering index (Han et al., 2007), grain size (Vandenberghe et al., 2004; Sun et al., 2006b, 2008) and terrestrial

mollusk records (Li et al., 2008), all suggest that significant East Asian monsoon oscillations might have been persistent since the Miocene (see a review by Guo et al., 2008).

During the interval 4.2–2.75 Ma, orbital-scale variability of both winter and summer monsoons was damped, implying decreased seasonality and/or glacial–interglacial contrasts. Filtered components of the two proxies at three primary orbital bands (i.e., 100-, 41-, and 23-kyr) reveal that damped amplitude is most likely related to weakened variability of the precession response, rather than to changes in variability at the 41- or 100-kyr components (Fig. 5). The precession component of the marine $\delta^{18}\text{O}$ record displays a uniform amplitude variation prior to 2.75 Ma, while the obliquity component of the $\delta^{18}\text{O}$ record slightly increased around 4.2 Ma, implying that ice volume change is not the factor driving the damped variability of the wind and precipitation regimes. Notably, the obliquity components of the marine $\delta^{18}\text{O}$ record, SMI and QGS exhibit a similar amplitude variation, damped prior to 2.75 Ma and enhanced afterward, suggesting that Northern Hemisphere ice sheets might have played a key role in modulating the obliquity-scale variability of the East Asian monsoon variability after 2.75 Ma.

Our monsoon records offer a plausible evidence for how Tibetan uplift might trigger the onset of the Northern Hemisphere glaciation. First, enhanced uplift of the Himalaya–Tibetan Plateau around 4.5–2.6 Ma (e.g., Li et al., 1997; Zheng et al., 2000, 2004) may have resulted in increased summer monsoon circulation (as evidenced by the long-term increasing SMI trend) and therefore greater moisture transportation to the high-latitude Northern Hemisphere (An et al., 2001). At the same time, tectonic changes may have brought the global climate to a critical state, with global temperatures elevated by as much as 3 °C with respect to modern values (Ravelo et al., 2004). This period is referred to as the mid-Pliocene climatic optimum and characterized by a permanent El Niño-like state (Wara et al., 2005) and by a damped response of monsoon seasonality to orbital forcing as inferred by our monsoon proxies. As a result, more moisture transport due to strengthened summer monsoon circulation may have led to increased ice accumulation in the high-latitude Northern Hemisphere, whilst decreased seasonality would permit more ice accumulation during the wet winter season and prohibit ice melting during the following cold-summer season (Maslin et al., 1998). Thus, a combined tectonic–orbital forcing, as suggested by Prell and Kutzbach (1992) and Maslin et al. (1998), may have led to the initiation of a major Northern Hemisphere glaciation.

After 2.75 Ma, large amplitude variations in the QGS and SMI suggest that the East Asian summer and winter monsoon circulation were dynamically linked to orbitally-induced changes in solar radiation and global ice volume (An et al., 1991a; Liu and Ding, 1993; Ding et al., 1995). The remarkable shift in the East Asian monsoon circulation around 2.75 Ma can most likely be attributed to internal forcing (e.g., ice volume change), since no amplitude shift is evident in the three orbital parameters (Fig. 5). Filtered components of the SMI and QGS exhibit a large amplitude shift at the 41- and 23-kyr bands, consistent with filtered signals of the $\delta^{18}\text{O}$ record. We know that ice volume change shows a corresponding shift around 2.75 Ma, indicating the onset of the Northern Hemisphere glaciation. Thus, development of the Northern Hemisphere glaciation might have played a key role in driving the remarkable shift of the Asian monsoon circulations (Ding et al., 2000; An et al., 2001), particularly in amplifying the monsoon response to obliquity forcing (Fig. 5B).

The filtered components indicate increased amplitude of the 100-kyr component for the $\delta^{18}\text{O}$ record around 1.25 Ma and for the two monsoon proxies around 0.6 Ma; however, the amplitude of the 41- and 23-kyr components of both monsoon proxies and $\delta^{18}\text{O}$ does not change significantly after 2.75 Ma (Fig. 5). Since ice sheets in the high-latitude northern hemisphere were fully developed around 1.25 Ma and subsequently experienced large-amplitude waxing and waning (Zachos et al., 2001; Raymo et al., 2006), we argue that the large shift

around 1.25 Ma and increased monsoon variability afterwards are strongly coupled with the full development of northern hemisphere ice sheets, particularly over the past 0.8 Ma (Ding et al., 1995). Moreover, a prominent change in the SMI around 0.6 Ma reveals that the summer monsoon intensity was significantly intensified during the subsequent interglacials. This remarkable strengthening of the summer monsoon intensity was previously attributed to the tectonic uplift of the Tibetan Plateau (An et al., 1990; Xiao and An, 1999). More recently, the strong asymmetry of bi-hemispheric climates (i.e., a significantly cooler Southern Hemisphere vs. an unusually warmer Northern Hemisphere) was suggested to be a prominent factor triggering this sudden strengthening of the summer monsoon intensity (Guo et al., 2009).

5. Orbital-scale coupling of the wind and precipitation variability

It is well known that during the Pleistocene strong winter monsoon (wind) and summer monsoon (precipitation) were generally 180° out-of-phase at glacial–interglacial scales, expressed by the alternating dominance between a strong summer monsoon during interglacials and a strong winter monsoon during glacials (An, 2000). Phase relations for the Pliocene interval, however, indicate that strong winter and summer monsoons were in-phase with one and another (Clemens et al., 2008), a finding that is clear in our data as well. Since magnetic susceptibility of the Red Clay is less sensitive to orbital-scale monsoon variations, the implications of the in-phase relationship between the summer and winter monsoons need further investigation. Here we evaluate the phase relationship between the precipitation (SMI) and wind (QGS) using two approaches. First, the time-varying correlation coefficient between the QGS and SMI was calculated using a 200-kyr window with 20-kyr time steps (e.g., 0–200 ka, 20–220 ka, etc.). Second, evolutive cross-spectral analysis of the QGS with SMI was performed using a 500-kyr window with 100-kyr time steps (e.g., 0–500 ka, 100–600 ka, etc.) to ascertain the phase relationship between these two proxies.

The time-varying correlation coefficient between the QGS and SMI of the Red Clay deposits differs significantly from that of the

loess–paleosol sequence. A rampfit function regression to the correlation coefficient reveals a gradual shift in the relationship between QGS and SMI from 3.15 to 2.75 Ma. Prior to 3.15 Ma, positive coefficient values indicate an almost in-phase relationship between the wind strength and precipitation, whereas after 2.75 Ma negative values indicate that strong wind strength is associated with weak precipitation (Fig. 6B). For the interval 3.15 to 2.75 Ma, the coefficient values reveal a transitional pattern of the coupling mode between wind strength and precipitation. Cross-spectral results reveal that the power spectrum and coherence between these two records are identical after 2.75 Ma, characterized by dominant peaks and strong coherency at the 100-, 41-, 23- and 19-kyr bands. Prior to 3.15 Ma, spectral peaks and high coherence are identified at the obliquity and precession bands. Evolutive cross-spectral results between these two proxies show that phase relations changed from nearly in-phase prior to 3.15 Ma to 180° out-of-phase after 2.75 Ma at the obliquity and precession bands. Both rampfit regression fit and cross-spectral results suggest a transition of orbital-scale coupling between winter wind strength (QGS) and summer precipitation (SMI) occurred around 3.15–2.75 Ma. We explain next the paleoclimatic implications of this remarkable shift and the possible driving mechanisms.

Since grain size reflects mainly surface processes (An et al., 1991b; Xiao et al., 1992; Sun et al., 2006b), we prefer to consider the QGS as a weathering-resistant and reliable indicator of wind strength. As suggested by previous studies (Porter and An, 1995; Xiao et al., 1995; Sun et al., 2006a), orbital-to-millennial scale QGS variations are closely linked to changes in Northern Hemisphere high-latitude climate. In contrast, the SMI is strongly related to more complex post-depositional processes such as leaching, evaporation, and pedogenesis; all these processes are dependant on both precipitation and temperature. An out-of-phase relationship between the SMI and QGS since the Pleistocene suggests that strong wind and heavy precipitation occurred at opposite times within an orbital cycle. During warm and wet interglacials, high SMI indicates strong leaching and pedogenesis induced by heavy precipitation, whilst low QGS

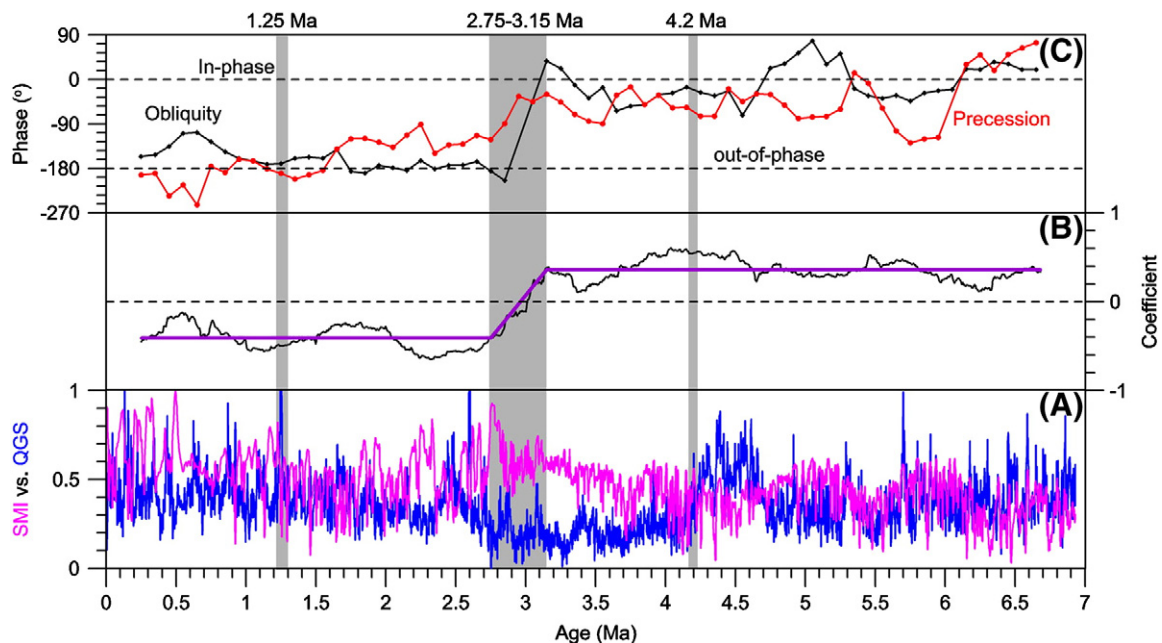


Fig. 6. (A) Variations of QGS (blue) and SMI (pink), (B) Time-varying correlation between QGS and SMI and rampfit function fit (heavy line) to the correlation coefficient, (C) Evolutive cross-spectral phase results for QGS and SMI at the obliquity and precession bands. These phase relationships are based on data from the same samples and are thus independent of the absolute chronology. 0° phase indicates that strong wind and precipitation are in-phase whereas 180° phase indicates that strong wind and precipitation are out-of-phase. Rampfit regression and cross-spectral analysis were performed using the RAMPFIT software (Mudelsee, 2000) and Arand software (Howell et al., 2006), respectively.

corresponds to relatively weak wind strength. During cold and dry periods, the leaching and pedogenic intensities would be significantly weakened due to limited precipitation, whereas the wind intensity was significantly strengthened.

In contrast, during the late Miocene and Pliocene high SMI (driven predominantly by the carbonate content change) coincides with coarser QGS, indicating that more effective precipitation occurred in cold times. As mentioned above, the SMI variation (particularly carbonate content) is controlled by the competing effects of precipitation-induced leaching and temperature-induced evaporation. If the SMI in the Red Clay formation was controlled mainly by precipitation-induced leaching, similar to the overlying loess–paleosol sequence, we can infer an in-phase relationship between strong wind and heavy precipitation prior to 2.75 Ma. This inference confirms previous findings based on the evolutive phase results for the magnetic susceptibility and QGS records, implying that heavy precipitation occurred during the winter season at obliquity minima (Clemens et al., 2008). Alternatively, if the SMI was related to temperature-controlled processes such as evaporation, higher SMI indicates less production of secondary carbonate due to weak evaporation under cold conditions. We then attribute the in-phase relationship between these two proxies prior to 3.15 Ma to the seasonal temperature difference. In either case, opposite phase relationships between the QGS and SMI indicate that the wind and precipitation coupling during the late Miocene to Pliocene was different from that of the Pleistocene glacial–interglacial cycles.

To ascertain the long-term evolution of the summer monsoon during the Pliocene to early Miocene, much research has focused on the types and concentrations of magnetic minerals that are strongly linked with post-depositional weathering and pedogenic processes. The Red Clay is characterized by a lower minimum magnetic susceptibility and by a relatively larger maximum superparamagnetic grain content than in the Quaternary loess–paleosol sequence, suggesting that magnetic minerals within the two units are different (Liu et al., 2003). Free Fe_2O_3 concentrations (mainly hematite and goethite) in the Red Clay exhibit large-amplitude oscillations with relatively high values compared to the overlying loess–paleosol sequences (Ding et al., 2001a), consistent with the relatively high hematite content and the ratio of hematite to goethite in the Red Clay formation relative to the overlying loess and paleosol layers (Ji et al., 2006). This magnetic evidence suggests that the late Neogene Red Clay formation was formed under relatively warmer and drier climatic conditions relative to those of the Pleistocene interglacials.

The 2.75-Ma climate shift is a global phenomenon although particularly dramatic in the monsoon-affected regions. For example, marine records of African climate document a shift toward more arid conditions and increased variability after 2.8 Ma (Tiedemann et al., 1994; deMenocal, 1995), though this shift was not supported by a statistical reassessment of previously published dust flux data (Trauth et al., 2009). The phase relationship between the Asian monsoon proxies and orbital forcing shifts significantly at both obliquity and precession bands around 2.7 Ma (Clemens et al., 1996, 2008). Our SMI variation reveals a relatively weak summer monsoon prior to 4.2 Ma followed by a continued intensification of the East Asian summer monsoon until 2.75 Ma, consistent with monsoon precipitation variability inferred from the carbonate isotope record of the loess and Red Clay sequences (An et al., 2005) and the hematite/goethite ratio of South China Sea sediments (Zhang et al., 2008). At the same time, the benthic $\delta^{18}\text{O}$ record reveals that the global temperature was about 3 °C warmer during the Pliocene compared to the Pleistocene (Zachos et al., 2001; Ravelo et al., 2004). Considering the effects of increasing precipitation and decreasing temperature together, we propose that prior to 2.75 Ma monsoonal circulation was characterized by alternating dominance of

windy-wet winter monsoons and warm-dry summer monsoons, with windy and wet winter seasons associated with low obliquity intervals. After 2.75 Ma, when the effects of NH ice volume came to dominate the system, glacial intervals (cold and dry) were characterized by strong winter monsoons and weak summer monsoons whereas interglacial intervals (warm and humid) were characterized by the opposite, weak winter monsoons and strong summer monsoons.

6. Conclusions

The East Asian monsoon oscillations over the past 7 Ma were reconstructed using variations of magnetic susceptibility, carbonate content and quartz grain size generated from a loess–paleosol and Red Clay sequence on the Chinese Loess Plateau. Our results suggest that both wind (QGS) and precipitation (SMI) exhibited significant orbital-scale variations prior to 4.2 Ma, inconsistent with previous suggestions based upon bulk grain size and magnetic susceptibility. Between 4.2 and 2.75 Ma, although the long-term trend of the summer monsoon was strengthened, the variability of both wind and precipitation was slightly damped. After 2.75 Ma, both wind and precipitation show large-amplitude glacial–interglacial variations, superimposed on a remarkable strengthening of the glacial winter monsoon and interglacial summer monsoon around 1.25 Ma. Variations in the QGS and SMI suggest that the coupling mode between the wind and precipitation during the late Miocene and Pliocene differed significantly from that of the Pleistocene glacial–interglacial cycles. Before 3.15 Ma, strong winter winds were coupled with heavy summer precipitation, whereas after 2.75 Ma strong winter winds were coupled with weak summer precipitation. The remarkable shift of the wind and precipitation coupling around 3.15–2.75 Ma suggests that the response of East Asian monsoon circulation to the orbital forcing was modulated greatly by the phased uplift of the Himalaya–Tibetan Plateau and the simultaneous expansion of the northern hemisphere ice sheets during the late Pliocene.

Acknowledgments

We thank Peter deMenocal and three anonymous reviewers for their valuable comments on this manuscript. This work was supported by the National Basic Research Program of China (no. 2010CB833403) and the “One-hundred Talents” program of the Chinese Academy of Sciences.

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